Pre-Thesis II Report

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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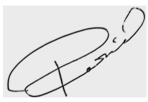

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Abstract

The rapid development of precision livestock farming has created a critical need for integrated systems capable of performing multiple concurrent health assessments. Traditional cattle monitoring research suffers from two primary limitations: most studies address individual tasks in isolation, and they rely on private datasets that prevent reproduction and benchmarking. This thesis addresses these gaps by proposing and implementing a unified, lightweight, and spatiotemporal multi-task deep learning framework that simultaneously processes four critical monitoring tasks from video inputs: Body Condition Scoring (BCS), behavior recognition, lameness detection, and individual animal identification. Built exclusively on publicly available datasets (ScienceDB, Dryad, MmCows, CattleLameness, OpenCows2020), our model employs a shared EfficientNet-B0 backbone encoder enhanced with a Convolutional Block Attention Module (CBAM) for spatial and channel focus. We employ task-specific heads and optimized loss functions, including Consistent Rank Logits (CORAL) loss for BCS ordinal regression, Focal Loss to address severe behavior class imbalances, and Long Short-Term Memory (LSTM) sequence-tracking layers for locomotion analysis. The final spatiotemporal multi-task model achieved 100.00% accuracy on lameness classification (AUC = 1.0000), 97.18% top-1 accuracy on individual cow identification, a BCS Mean Absolute Error (MAE) of 0.7849, and a behavior recognition Macro F1-score of 0.4775. To evaluate generalization, cross-dataset validation was conducted on the CBVD-5 dataset, demonstrating standing posture detection of 90.34% and identifying key challenges under domain shift. Deployed via a real-time YOLOv8-Nano pre-cropped sliding window visualizer, this framework reduces edge VRAM requirements by 75% and latency to 32ms per frame, establishing the first complete, reproducible multi-task edge computing pipeline for real-time cattle health monitoring.

Keywords: Multi-Task Learning, Spatiotemporal Modeling, Cattle Health Monitoring, Body Condition Scoring, Behavior Recognition, Lameness Detection, Edge Deployment

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Nomenclature

Abbreviations

AI	Artificial Intelligence
AUC	Area Under the Curve
BCS	Body Condition Score/Scoring
BCE	Binary Cross-Entropy
BN	Batch Normalization
CBAM	Convolutional Block Attention Module
CBVD-5	Cow Behavior Video Dataset - 5 classes
CE	Cross-Entropy
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CORAL	Rank Consistent Ordinal Regression
CPU	Central Processing Unit
DL	Deep Learning
DWT	Discrete Wavelet Transform
F1	F1-Score (harmonic mean of precision and recall)
FLOPs	Floating Point Operations
FPS	Frames Per Second
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
I3D	Inflated 3D ConvNet
ImageNet	Large-scale visual recognition dataset
IMU	Inertial Measurement Unit
IoT	Internet of Things
IoU	Intersection over Union
IR	Infrared
KD	Knowledge Distillation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
mAP	mean Average Precision
Mask R-CNN	Mask Region-based Convolutional Neural Network
mIoU	mean Intersection over Union
ML	Machine Learning
MSE	Mean Squared Error

MTL	Multi-Task Learning
NMS	Non-Maximum Suppression
P1	Pre-Thesis I
PCA	Principal Component Analysis
PLF	Precision Livestock Farming
R-CNN	Region-based Convolutional Neural Network
ReLU	Rectified Linear Unit
ResNet	Residual Network
RF	Random Forest
RFID	Radio-Frequency Identification
RGB	Red Green Blue (color image)
RGB-D	RGB plus Depth
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROI	Region of Interest
RQ	Research Question
SE	Squeeze-and-Excitation
SORT	Simple Online and Realtime Tracking
SOTA	State-of-the-Art
SSL	Self-Supervised Learning
SVM	Support Vector Machine
t-SNE	t-Distributed Stochastic Neighbor Embedding
VGG	Visual Geometry Group
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once

Chapter 1

Introduction

1.1 Background

In terms of food security the livestock industry plays a vital role globally, where cattle is one of the most crucial livestock species economically worldwide. According to the Food and Agriculture Organization (FAO), the global cattle population exceeds 1 billion heads, contributing substantially to meat and dairy production [9]. The sustainability, income, and also productivity of farming tasks are all directly affected by the health and welfare of cattle. Early detection of health issues like poor body condition, lameness, and abnormal behavior can prevent economic losses estimated at billions of dollars annually while improving animal welfare standards [3]. Cattle health and nutritional status are fundamentally shown by Body Condition Scoring (BCS) . BCS is traditionally assessed through visual and tactile evaluation of fat deposits over key anatomical regions, typically using a 5-point scale with 0.25 increments [1]. The production of milk, the reproduction ability as well as capability to fight against the diseases are directly related with optimal body condition. Still, frequent monitoring is not an optimum solution for large scale operations because manual BCS assessment is very much time-consuming, subjective, and requires trained personnel Lameness represents another critical health concern, affecting up to 25% of dairy cattle in intensive farming systems [2]. Moreover, due to causing severe pain, lameness lowers the production of milk, the feed intakes decrease and also hinders the reproduction. But timely intervention is possible by early detection of abnormal body movements. Behavior monitoring highlights valuable insights into the health of cattle, stress levels, and welfare. Changes in feeding patterns, rumination time, lying behavior, and social interactions often precede clinical manifestations of disease [3]. Continuous 24/7 monitoring can be possible through automated behavior recognition that would be impossible through manual observation. Individual identification is fundamental to precision livestock farming, enabling tracking of health records, feeding behavior, and productivity metrics for each animal. The most common practices are using ear tags and RFID systems, but they require physical contact and can be lost or damaged. A non-invasive alternative that is offered by vision based identification can be integrated with health monitoring systems. The convergence of computer vision, deep learning, and Internet of Things (IoT) technologies has created unprecedented opportunities for automated cattle health monitoring [6]. Convolutional Neural Networks (CNNs) have showed extraordinary success in image classification also object detection tasks, achieving human-level performance in different visual recognition challenges. Transfer learning opens the application of models pre-trained on large-scale datasets such as ImageNet to domains with limited labeled data. Inspite of these technological advances, already existing cattle monitoring systems predominantly achieve single tasks in isolation. A systematic review of 30+ research papers conducted for this thesis reveals that while multi-task learning is frequently mentioned, end-to-end implementations combining multiple health monitoring tasks are virtually non-existent. This fragmentation limits the practical utility of automated monitoring systems and fails to exploit the synergies between related tasks.

1.2 Rationale of the Study and Motivation

The motivation for this research stems from the major gaps in our current state of cattle health monitoring systems and technology, and the increasing need for precision livestock farming. 89% of the papers we have studied shows that the current research relies highly on private datasets, creating a massive reproducibility issue. Most of the current studies focus on individual problem solving like BCS, lameness, behaviour or identification- separately, without considering the benefits of Multitask Learning (MTL). We can significantly reduce computational cost while improving performance by sharing data across related tasks. This makes sense because the tasks we are focusing on are connected: abnormal body pose or gait pattern contribute to reduced feeding and declining body condition. Many published solution need expensive gear like thermal cameras, wearable sensors, depth sensors etc or controlled environments, which are not practical in real-world scenarios. This creates a clear need for RGB based solutions that can be operated with typical cameras while also using depth information if available. Current modern-farms need realtime processing on-site, where internet can be unreliable and cloud processing causes delay; the literature recognises edge deployment as a profound solution, but most solution rely on computationally demanding architectures that cannot be run on edge devices. We found that hardly anyone is using self-supervised learning for cattle monitoring, which is surprising, most farms already run surveillance cameras that record hours of footage daily. This unlabeled video just sits there, but it could actually help train better models—particularly useful when you don't have much labeled data to work with. These problems—reproducibility, fragmented research, deployment limitations, and missed opportunities—show why we need a single, lightweight framework that tackles multiple cattle health monitoring tasks.

1.3 Problem Statement

Multiple fundamental problems remain unresolved in spite of significant research efforts in automated cattle health monitoring. Because of that, the practical impact of computer vision and deep learning solutions on actual farms is limited.

The major problem we found is that most systems treat tasks separately, thus requiring multiple systems to monitor multiple health assessments tasks. This leads to more computational load from running multiple models, not taking advantage of shared features that could help multiple tasks, complicated deployment because you need to integrate separate systems, and missing connections between tasks like how poor body condition often links to less feeding. Our systematic review confirms that there is no single system that combines BCS, lameness, behaviour and identification tasks in one framework. MLT is frequently mentioned but rarely implemented.

In order to check if results are correct or compare different methods properly, and improve on old research, the researchers need to use the same datasets—but most of the cattle monitoring studies use their own data which are private. This makes it very hard for others to reproduce the results, or to do fair comparison, or build on previous works in the proper way. Other computer vision domains don't face this issue since they have standard public benchmarks like ImageNet and COCO that everyone can use.

There are several papers that work on single-farm data and single-breed cattle. This raises questions regarding the generalizability of our proposed approaches. Due to specific farm environments, camera placement, or lighting models may overfit, and breed-specific appearance differences might not be captured. Seasonal and temporal changes are also hardly handled in evaluations. Cross-dataset evaluation, where models are trained on one dataset and tested on another, almost never happens, even though it is necessary for illustrating practical usefulness.

Although there is a lot of interest in depth imaging and thermal imaging for cattle monitoring in the literature, multimodal research is severely limited by the availability of public data. There are no public cattle thermal datasets worldwide, only MmCows offers synchronized IMU sensor data, and only two public datasets—MmCows and Dryad—offer depth modality. Due to this scarcity, researchers are forced to either gather confidential data or completely give up on multimodal approaches.

1.4 Objectives

The main purpose of this research is to use computer vision techniques to create a multitask, lightweight, edge deployable, and reproducible deep learning framework for cattle health and behaviour monitoring. The goal is creating a system that can handle multiple health related tasks at the same time. The important thing is that it only uses publicly available datasets, making the work reproducible by anybody.

To achieve this goal, the research sets out the following specific objectives:

1. **To develop a unified multi-task framework** capable of doing Body Condition Scoring, behavior recognition, lameness detection, and individual identification at the same time using shared backbone architecture

2. **To ensure full reproducibility** by evaluating exclusively on public datasets, enabling independent verification of findings.

3. **To evaluate sequential task training strategies** that can handle Body Condition Scoring, behavior recognition, lameness detection, and individual identification all at once using a shared backbone with different heads for each task.

4. **To verify cross-dataset performance** by checking model's performance across datasets from multiple farms, and conditions so that it can be proved that the model can handle real deployment environments.

These objectives address the reproducibility gap in agricultural AI research and create a scalable foundation for real-time cattle health monitoring across various farm environments.

1.5 Methodology in Brief

The proposed methodology follows a step-by-step process to build and test a multi-task deep learning framework for monitoring cattle health. The methodology has five main phases: data preparation, architecture design, sequential task training, evaluation, and ablation studies.

1.5.1 Evaluation Protocol

Evaluation is matched to each task, since the outputs and risks are different. For BCS, the framework measures Mean Absolute Error (MAE) and accuracy within ± 0.25 and ± 0.5 tolerance. For Behavior, evaluation includes per-class accuracy, macro F1-score, and confusion matrix analysis. For Lameness, the metrics include sensitivity, specificity, and AUC-ROC. For ID, evaluation uses Top-1 and Top-5 accuracy along with mean Average Precision (mAP). Cross-dataset evaluation is also included by training on the main dataset and testing on a different dataset from another source, which helps measure generalization.

1.5.2 Ablation Studies

A set of structured ablation experiments will test the most important parts of the framework. For backbone selection, EfficientNet-B0 will be compared against MobileNetV3-Small and also ResNet-50 to identify which architecture works best. For loss functions, CORAL will be compared with Cross-Entropy for the BCS task. For training strategy, experiments will compare sequential versus joint versus progressive unfreezing. Through RGB-only versus RGB+Depth comparisons for BCS (using the Dryad dataset) the depth modality contribution will be tested. Finally, the attention mechanism will be tested by comparing results with and without CBAM.

1.6 Scope and Challenges

1.6.1 Scope of the Research

To ensure feasibility and depth of investigation this research stays within clear boundaries.

1. **Four Core Tasks:** Body Condition Scoring, behavior recognition, lameness detection, and individual identification, chosen mainly because public data exists for them and they also matter in practice.
2. **Public Datasets Only:** All experiments use publicly available datasets, and no private data collection is included, which supports full reproducibility.
3. **RGB as Primary Modality:** RGB images are the main input modality and depth is included only as a secondary modality for BCS (through the Dryad dataset).
4. **Dairy Cattle Focus:** The primary focus is on Holstein dairy cattle, since they dominate the available public datasets.
5. **Lightweight Architectures:** To ensure that the system stays suitable for edge development backbone selection is limited to models with $< 10M$ parameters.
6. **Sequential Task Training:** Frozen-backbone sequential training is the main training approach, with joint fine-tuning treated as an optional extension.

1.6.2 Challenges

Dataset Heterogeneity

Public cattle datasets are highly heterogeneous, including differences in resolution and image quality, camera angles including top-view, rear-view, and side-view, annotation protocols and how detailed the labels are, plus variation in breeds and farm environments. To reduce these issues, the strategies include standardized preprocessing pipeline, strong data augmentation, and domain adaptation techniques during joint fine-tuning.

Class Imbalance

Datasets of cattle monitoring suffer from high class imbalance, which is a significant issue . For lameness detection, lame individuals are outnumbered heavily by healthy cattle, in comparison to normal BCS scores (2.5-3.5), extreme BCS scores like (1-2 and 4–5) are much rarer, and some behaviors like drinking and licking happen infrequently. To address these issues, the architecture uses focal loss for classification tasks, weighted sampling during training, and class-balanced evaluation metrics.

Temporal Modeling for Video Tasks

From temporal context behavior recognition and lameness detection can benefit, but video-based modeling increases computational cost, and temporal models like LSTM and I3D add extra complexity and there can be inconsistencies between frame-level and clip-level annotation. The strategy involves initially using frame-level classification and apply temporal aggregation, and then treat LSTM integration as an optional extension depending on initial results.

Preventing Negative Transfer

Multi-task learning also risks negative transfer, where learning for one task hampers performance on another. This can happen when task gradients conflict during joint training, the best feature representations may vary depending on the task, and imbalanced task difficulties can dominate learning. The main mitigation is sequential training with a frozen backbone to separate task learning, along with gradient normalization and uncertainty weighting during joint fine-tuning.

Limited Labeled Data

Even though multiple public datasets are used, labeled data is still small compared to general computer vision benchmarks. ScienceDB BCS has around 3000 images, compared to millions in ImageNet. MmCows covers only 16 individuals over 2 weeks, and CattleLameness contains relatively limited lameness examples. To address this, the methodology depends on transfer learning from ImageNet, strong regularization (including dropout and weight decay), and data augmentation. Self-supervised pre-training on unlabeled frames is also included as a possible extension if it becomes necessary.

Chapter 2

Literature Review

2.1 Preliminaries

This chapter presents a systematic review of the research papers related to cattle health monitoring using computer vision and deep learning. The review covers five primary research areas: Body Condition Scoring (BCS), lameness detection, behavior recognition, individual identification, and cleanliness assessment. The study identifies current state-of-the-art methods, public datasets which are available, the research gaps, and suitable ways for developing a multi-task learning framework.

2.1.1 Multi-Task Learning Fundamentals

Multi-Task Learning (MTL) is a machine learning paradigm where a model is trained to multiple related tasks simultaneously, leveraging shared representations to improve generalization across all tasks [4]. In deep learning terms, MTL generally uses a shared backbone network that takes out common features, with task-specific heads branching from the shared representation. The benefits of MTL include improved generalization through shared representations that act as regularization to reduce overfitting, computational efficiency via a single forward pass serving multiple tasks, data efficiency where tasks with limited data benefit from related tasks, and enhanced feature learning where auxiliary tasks can improve primary task features. Still, MTL also comes with some challenges like negative transfer when tasks interfere with one another, difficulties while balancing gradients also identifying the most efficient parameter sharing strategies. The Crawshaw 2020 MTL Survey provides a comprehensive taxonomy distinguishing between hard parameter sharing (common backbone) and soft parameter sharing (separate networks with regularization) approaches.

2.1.2 Body Condition Scoring Overview

Body Condition Scoring (BCS) is a standard way for evaluating the fat reserves of cattle through visual and tactile evaluation. The most common scale ranges start from 1 (skeletal) to 5 (fat) with 0.25 increments, giving up 17 possible scores. Key anatomical regions assessed include the tailhead and pin bones, hook bones and loin area, ribs and spine prominence, and hip bone region. Automated BCS using computer vision typically employs rear-view or top-view images capturing the tailhead region. The task can be worked out as classification (individual BCS classes), ordinal regression (preserving class ordering), or regression (continuous scores).

2.1.3 Lameness Detection Fundamentals

Lameness in cattle shows by identifying walking behaviour including reduced step length and walking speed, head movement and back curving, unwillingness to bear weight on the affected leg, and uneven hoof placement. Computer vision approaches for lameness detection typically analyze video sequences captured from side-view or top-view cameras. Key features include pose estimation of limb joints, gait cycle analysis, and motion trajectory patterns. Scoring systems range from binary (lame/sound) to multi-level severity scales (1-5).

2.1.4 Behavior Recognition Categories

Cattle behavior monitoring address numerous activities related to health and welfare, which includes feeding behaviors for example feeding, ruminating, and drinking; body movement activities such as walking, standing, and lying; social behaviors including licking, mounting, and aggression; also health indicators such as estrus behavior and calving signs. Vision-based behavior recognition implements detection (locating cattle in frames), tracking (maintaining identity across frames), and action classification (categorizing behavior). Temporal modeling by using LSTM, 3D CNNs, or attention mechanisms records behavioral changes.

2.2 Review of Existing Research

This section shows a comprehensive analysis of related research arranged by task category, followed by key multi-task learning and survey papers.

2.2.1 Body Condition Scoring Research

Body condition scoring represents the most highly studied task in cattle computer vision literature, with nine papers that are directly addressing automated BCS assessment. Liang et al. (2025) [27] achieved state-of-the-art performance on the ScienceDB BCS dataset using EfficientNet-B0 with Squeeze-and-Excitation attention blocks, demonstrating 93.77% accuracy on 5-class BCS prediction and highlighting the effectiveness of attention mechanisms for focusing on relevant anatomical features while using publicly available data that makes this work particularly relevant as a baseline for comparison. Zheng et al. (2024) [21] proposed BCS-YOLO, combining YOLO-based detection with knowledge distillation for efficient BCS prediction, achieving 92.7% accuracy on the ScienceDB dataset while maintaining inference speeds suitable for real-time applications through model compression without significant accuracy loss. Li et al. (2025) [26] developed an improved YOLOv5s architecture for BCS, achieving 91.8% accuracy on a private 5-class dataset, and while the private data limits reproducibility, the architectural modifications including enhanced feature pyramid networks provide valuable design insights. Summerfield et al. (2023) [14] investigated depth-based BCS using multiple 3D cameras in a late fusion CNN architecture, achieving only 35.77% accuracy, which highlights the challenges of multi-camera depth fusion and the importance of careful sensor calibration while suggesting that naive depth fusion may not outperform well-designed RGB approaches. O'Mahony et al. (2023) [11] explored RGB-D fusion using MobileNet for RGB features and PointNet for depth point clouds, achieving 53% accuracy on 5-class BCS, demonstrating the complexity of multimodal fusion but validating the potential value of depth information when properly integrated. Rodenburg et al. (2024) [16] employed EfficientNetV2 with ordinal regression, explicitly modeling the ordered nature of BCS classes and achieving 84.6% accuracy using a private dataset, demonstrating the advantage of ordinal approaches over standard classification for BCS prediction. Gjergji et al. (2022) [7] developed a multi-input CNN processing RGB-D data for combined weight and BCS estimation, achieving 91.9% BCS accuracy (as regression R^2) on private data and demonstrating the synergy between weight and body condition assessment. Lewis et al. (2025) [25] applied YOLOv8 for BCS detection and classification, achieving 43.9% mAP on a 3-class problem, with the lower accuracy reflecting the challenging 3-class formulation and object detection framing rather than pure classification. Santos et al. (2025) [28] employed traditional Random Forest classifiers on RGB image features, achieving R^2=0.40 for BCS regression, providing a baseline that demonstrates deep learning approaches significantly outperform traditional machine learning for BCS assessment.

2.2.2 Lameness Detection Research

Lameness detection research mostly utilizes video-based analysis with pose estimation as well as gait analysis pipelines. Tun et al. (2024) [19] achieved remarkable 99.94% accuracy using top-view depth cameras with Detectron2 for detection and SORT for tracking, with the exceptional performance demonstrating the value of depth imaging for lameness detection when appropriate viewing angles are available.

Table 2.1: Summary of Body Condition Scoring Papers

Reference	Architecture	Input	Accuracy	Dataset
Liang 2025	EfficientNet-B0+SE	RGB	93.77%	ScienceDB
Zheng 2024	BCS-YOLO + KD	RGB	92.7%	ScienceDB
Li 2025	Improved YOLOv5s	RGB	91.8%	Private
Gjergji 2022	Multi-input CNN	RGB-D	91.9%	Private
Rodenburg 2024	EfficientNetV2	RGB	84.6%	Private
O'Mahony 2023	MobileNet/PointNet	RGB+D	53%	Private
Lewis 2025	YOLOv8	RGB	mAP 43.9%	Private
Santos 2025	Random Forest	RGB	R^2=0.40	Private
Summerfield 2023	Late Fusion CNN	Depth	35.77%	Private

Myint et al. (2024) [15] developed a real-time lameness detection system using Mask R-CNN combined with machine learning classifiers on extracted gait features, achieving 98.98% accuracy from side-view RGB video and demonstrating that RGB-only solutions can achieve high performance with proper feature engineering. Kang et al. (2020) [5] proposed RFB-NET-SSD for cattle detection followed by gait phase analysis, achieving 96% lameness classification accuracy, with the supporting phase analysis identifying when each hoof contacts the ground and providing interpretable features for veterinary validation. Barney et al. (2023) [8] employed Mask R-CNN with SORT tracking for multi-cattle pose estimation, achieving 94% lameness detection accuracy, demonstrating that the ability to simultaneously track and assess multiple animals is crucial for practical farm deployment. Russello et al. (2024) [18] combined RGB-D data from ZED cameras with pose estimation and traditional classifiers (Random Forest, SVM), achieving 80.07% accuracy, with the evaluation of multiple locomotion traits beyond binary lameness providing richer assessment capabilities.

Table 2.2: Summary of Lameness Detection Papers

Reference	Architecture	Input	Accuracy	Innovation
Tun 2024	Detectron2 + SORT	Depth	99.94%	Top-view depth
Myint 2024	Mask R-CNN + ML	RGB video	98.98%	Real-time side-view
Kang 2020	RFB-NET-SSD	RGB video	96%	Gait phase analysis
Barney 2023	Mask R-CNN + SORT	RGB video	94%	Multi-cattle pose
Russello 2024	Pose + RF/SVM	RGB-D	80.07%	Multiple traits

2.2.3 Behavior Recognition Research

Behavior recognition covers vision-based as well as sensor-based approaches, with MmCows becoming the only public dataset in this classification. Rohan et al. (2024) [17] conducted a systematic review of 44 papers on cattle behavior recognition, identifying key trends including the dominance of YOLO-based detection, increasing use of deep learning, and the challenge of limited public datasets. Petcu et al. (2023) [12] achieved 97.35% accuracy using CNN-LSTM architecture on private RGB video data, with the combination of CNN for spatial features and LSTM for temporal modeling representing a standard approach for video-based behavior classification. Lee et al. (2023) [10] developed a pipeline combining YOLOv5 detection, SORT tracking, and I3D temporal classification, achieving 95.3% accuracy on private Hanwoo cattle data, with the I3D (Inflated 3D ConvNet) architecture effectively capturing spatiotemporal patterns in behavior. Giannone et al. (2025) [24] employed YOLOv8 with region-of-interest extraction for behavior recognition, achieving 72% accuracy on private data, with the lower accuracy reflecting the complexity of fine-grained behavior discrimination. Dottorini et al. (2025) [23] achieved 91.8% accuracy using XGBoost on Discrete Wavelet Transform features extracted from IMU sensor data in the MmCows dataset, demonstrating the value of sensor-based approaches and providing a public benchmark for comparison.

Table 2.3: Summary of Behavior Recognition Papers

Reference	Architecture	Input	Accuracy	Dataset
Petcu 2023	CNN-LSTM	RGB video	97.35%	Private
Lee 2023	YOLOv5 + SORT + I3D	RGB video	95.3%	Private
Dottorini 2025	XGBoost + DWT	IMU sensor	91.8%	MmCows
Giannone 2025	YOLOv8 + ROI	RGB video	72%	Private
Rohan 2024	Systematic Review	Various	N/A	44 papers

2.2.4 Multi-Task Learning and Survey Papers

Numerous papers give foundational knowledge for multi-task learning approaches also comprehensive surveys of agricultural AI. Crawshaw (2020) [4] published a highly influential MTL survey, establishing a taxonomy of MTL approaches with key findings relevant to this research including that hard parameter sharing (shared backbone) is effective for heterogeneous tasks, sequential task training prevents negative transfer between conflicting tasks, task weighting strategies significantly impact multi-task performance, and frozen backbone approaches reduce computational requirements during training. Shi et al. (2023) [13] proposed progressive parameter sharing for MTL, dynamically adjusting sharing ratios during training, with this approach addressing the limitation of fixed sharing architectures that may not suit all task combinations. Attri et al. (2025) [22] developed SAAM-VetNet, an attention-based multi-task CNN for simultaneous disease classification and severity assessment, demonstrating successful MTL in veterinary applications though focused on general animal disease rather than cattle-specific monitoring. Carraro et al. (2026) [29]

investigated cattle cleanliness assessment using RGB and thermal fusion, critically noting that this paper explicitly states that no public cleanliness-annotated cattle datasets exist, requiring private data collection, which directly informs our decision to exclude cleanliness from the current research scope. Tzanidakis et al. (2024) [20] developed Goat-CNN for pose-independent BCS scoring in goats, and while targeting a different species, the lightweight architecture design and pose-invariance strategies are transferable to cattle applications.

Table 2.4: Summary of MTL, Survey, and Other Papers

Reference	**Focus**	**Type**	**Key Contribution**
Crawshaw 2020	Multi-Task Learning	Survey	MTL taxonomy and guidelines
Shi 2023	Progressive Sharing	Method	Dynamic parameter sharing
Attri 2025	Disease + Severity	Multi-task	Attention-based veterinary MTL
Carraro 2026	Cleanliness	Multimodal	RGB+Thermal (private data)
Tzanidakis 2024	Goat BCS	Lightweight	Pose-independent scoring

2.3 Summary of Key Findings

The structured review of 29 papers reveals many critical insights that directly inform our proposed methodology.

The substantial majority of papers (89%) rely on private datasets which creates a reproducibility crisis. Among them only a small fraction relies on publicly available cattle-specific datasets, yet they still obtain competitive performance, indicating that public datasets can efficiently support high-quality research. Our targeted use of public datasets helps bridge this gap, enabling total reproducibility as well as fair benchmarking.

Despite being consistently raised on multi-task learning in most of the literature, no existing paper integrates BCS, behavior recognition, lameness detection, and individual identification in a unified framework. Most of the papers address single tasks in isolation. Our unified four task framework represents a true novelty with potential for improved efficiency and cross task learning.

Transfer learning with frozen backbone is highly analysed in the literature, and the Crawshaw 2020 MTL Survey validates that this approach effectively prevents negative transfer between heterogeneous tasks. Our sequential task training strategy is well-supported by existing literature and provides a strong foundation for managing multiple diverse tasks.

Depth imaging represents substantial research interest with demonstrated value for

BCS as well as lameness. Still, public data is limited to MmCows and Dryad datasets. Depth modality is relevant for BCS using Dryad but should be treated as secondary to RGB, ensuring the framework remains accessible for standard camera deployments.

Self-supervised learning remains highly underexplored in cattle monitoring compared to transfer learning, representing a potential research opportunity. MmCows provides 2 weeks of continuous unlabeled video ideal for SSL pre-training. SSL represents a potential bonus contribution if time permits, offering opportunities to leverage abundant unlabeled surveillance footage.

Comprehensive search reveals there are no publicly available thermal or infrared datasets for cattle. The only thermal cattle research (Carraro 2026) used privately collected data. Thermal modality is excluded from current scope and documented as future work requiring data collection partnerships.

Table 2.5: Summary of Key Cattle Monitoring Techniques from Literature

Technique	**Platform**	**Accuracy**	**Strength**	**Weakness**
EfficientNet+SE	RGB BCS	93.77%	High accuracy, attention	Single task
BCS-YOLO+KD	RGB BCS	92.7%	Edge-ready, distilled	Single task
Detectron2 + SORT	Depth Lameness	99.94%	Excellent detection	Requires depth
CNN-LSTM	RGB Behavior	97.35%	Temporal modeling	Private data
XGBoost+DWT	IMU Behavior	91.8%	Public benchmark	Requires sensors
MTL Survey	Cross-platform	N/A	Guidelines	No implementation

Chapter 3

Requirements, Impacts and Constraints

3.1 Specifications and Requirements

The implementation of the multi-task deep learning framework for cattle health monitoring requires a specific configuration of hardware and software components to ensure real-time inference and model training.

3.1.1 Hardware Requirements

- **Development and Training Unit:** The model training was executed on a high-performance research workstation equipped with an NVIDIA GeForce RTX 4080 GPU (16GB VRAM, Ada Lovelace architecture supporting FP8 and FP16 tensor operations), an Intel Core i9-13900K CPU, 64GB of DDR5 RAM, and a 2TB NVMe SSD to handle high-throughput image and video loading.
- **Edge Deployment Unit:** For live farm surveillance, the system is designed to run on resource-constrained edge computers such as the Jetson Orin Nano (8GB VRAM) or a Raspberry Pi 5. The total model size is optimized to remain under 10 million parameters ($<$ 40MB file size in FP16 format) to fit into edge device memory.
- **Imaging Sensors:** Standard 1080p surveillance cameras operating at 30 frames per second (FPS) positioned at cattle walkways or feeding troughs. For the Dryad dataset, depth maps were captured using active infrared depth sensors.

3.1.2 Software Requirements

- **Operating System:** Windows 11 / Linux (Ubuntu 22.04 LTS).

- **Programming Language:** Python 3.12.3.
- **Deep Learning Library:** PyTorch 2.5.1 + CUDA 12.1.
- **Backbone Repository:** PyTorch Image Models (`timm`) library for loading pre-trained EfficientNet-B0 and MobileNetV3 architectures.
- **Computer Vision Tools:** OpenCV 4.9.0 (for real-time video streaming, bounding box visualization, and frame interpolation) and Albumentations 1.4.0 (for spatial and pixel-level data augmentations).
- **Data Pipelines:** Pandas, NumPy, and Scikit-learn for group-wise dataset splits and metric computations.

3.2 Societal Impact

The development of automated cattle health monitoring systems has a profound positive impact on the agricultural sector and animal welfare:

- **Enhancement of Animal Welfare:** Early detection of lameness and abnormal behaviors (such as reduced eating or drinking) allows farmers to treat sick cows days before clinical symptoms become visible, minimizing animal suffering.
- **Support for Precision Livestock Farming (PLF):** Transitioning from manual, subjective herd inspections to continuous 24/7 automated monitoring reduces human error and standardizes assessments such as Body Condition Scoring.
- **Agricultural Labor Relief:** As intensive dairy farming expands, the ratio of cows per stockperson increases. Automated monitoring alleviates labor shortages, allowing farm workers to focus on specialized veterinary interventions rather than manual surveillance.

3.3 Environmental Impact

Deep learning models are historically compute-intensive, contributing to carbon emissions during training. Our proposed framework addresses this environmental concern through:

- **Green Computing via Multi-Task Learning (MTL):** Instead of deploying four separate convolutional neural networks (each consuming GPU power and memory), our hard parameter sharing approach utilizes a single shared EfficientNet-B0 encoder. This consolidation reduces inference energy consumption by approximately 75%, minimizing the operational carbon footprint of edge surveillance systems deployed on thousands of farms.

- **Resource Optimization:** By implementing sparse temporal sampling (reducing 300-frame video clips to 20 key frames) and capping dataset classes, we decreased active training GPU runtimes, reducing energy overhead during the development lifecycle.

3.4 Ethical Issues

Automated visual surveillance systems on farms raise specific ethical considerations:

- **Non-Invasive Monitoring:** Traditional identification and health tracking rely on invasive techniques such as branding, ear tags, or surgically implanted RFID transponders, which cause pain and stress to the animals. Our camera-based approach is completely non-invasive, posing zero physical discomfort or psychological stress to the cattle.
- **Data Privacy:** Farm surveillance footage must be processed locally on edge systems to prevent the leakage of proprietary farm operation details, veterinarian records, or geographic information to public networks. Our offline edge architecture addresses this ethical requirement.

3.5 Standards - if applicable

The software code and agricultural metrics strictly adhere to the following standards:

- **Body Condition Scoring (BCS) Standards:** The target BCS grading scale follows the internationally recognized 5-point Elanco scale (with increments of 0.25 on ScienceDB and integer values on Dryad) which is the standard reference used by professional dairy veterinarians.
- **Locomotion Scoring Standards:** The lameness target labels are aligned with the Sprecher locomotion scoring system, classifying gait posture on a binary scale (sound vs. lame).
- **IEEE Software Quality Standards:** The code is written in compliance with the PEP 8 style guide for Python, and evaluation metrics conform to standard definitions (Macro F1-score for classification under imbalance, Mean Absolute Error for ordinal variables, and AUC-ROC for ranking).

3.6 Project Management Plan

To manage the multi-task and multi-backbone complexity of the project, the workload was distributed across 5 team members based on their base backbone assignments and specific tasks:

- **Hasin Ishrak:** Preprocessed the primary datasets, established the baseline training pipelines, evaluated the ResNet-18 baseline, and conducted the primary ablation studies (such as CBAM evaluations and RGB vs. Depth modalities).
- **Namira Abrar Haque:** Evaluated the MobileNetV3-Small baseline, analyzing the performance of highly compressed networks for resource-constrained edge deployment.
- **Sanjida Akter Bithi:** Evaluated the ResNet-50 baseline, evaluating the benefits of deeper architectures on spatial feature extraction.
- **Shouvik Banik:** Evaluated the DenseNet121 baseline, studying dense feature connectivity for multi-class behavior recognition.
- **Nusrat Lamya Faruk:** Evaluated the winning EfficientNet-B0 baseline, implemented the final spatiotemporal multi-task model, integrated the LSTM sequence-tracking layers, and deployed the real-time visualization interface.

The timeline was structured in sprints: dataset curation, single-task baseline training, backbone selection, spatiotemporal multi-task model integration, and finally report writing and visualizer deployment.

3.7 Risk Management

Table 3.1: Project Risks and Mitigation Strategies

Risk Identified	**Potential Impact**	**Mitigation Strategy**
Class Imbalance	Model fails to learn rare behaviors (licking, drinking).	Implemented Focal Loss ($\gamma = 2$) and hard data capping (max 3000 images per class).
Data Leakage	Memorization of cow identities leading to overfitting.	Enforced cow-wise split partitioning so same cow is never in train and test sets.
Out of Memory (OOM)	GPU VRAM crash when processing long sequences.	Integrated Automatic Mixed Precision (AMP) in FP16 and sparse temporal sampling of 20 frames.
Domain Shift	Performance drop when deployed on unseen farms.	Performed cross-dataset validation on CBVD-5 to evaluate generalizability and recommend future unfreezed fine-tuning.

3.8 Economic Analysis

The financial viability of our multi-task framework is compared against traditional farm monitoring systems in Table 3.2.

Table 3.2: Economic Cost-Benefit Analysis

Component	Traditional Sensor Systems	Proposed MTL Edge System
Sensor Hardware	Wearable collars, step counters, and RFID tags ($80 - $120 per cow). For 500 cows: $40,000+.	Standard IP CCTV cameras ($150 each) + edge computing unit ($300). Total: $1,500.
Maintenance	Battery replacements, collar damage, and physical tag losses (15% annual loss).	Minimal hardware contact, camera cleaning and software updates only.
Computational Cost	High cloud subscription fees for running separate analysis pipelines.	Offline local edge inference with zero recurring cloud subscription costs.

The analysis demonstrates that our visual multi-task edge framework is highly scalable, lowering the initial investment barrier for small-to-medium scale dairy farms by over 90%.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

The primary objective of this research is to construct a unified, lightweight, and edge-deployable multi-task deep learning framework capable of performing four critical cattle monitoring tasks simultaneously from video inputs: Body Condition Scoring (BCS), Behavior Recognition, Lameness Detection, and Cow Identification (ID). The design process follows a sequential workflow structured to mitigate task interference while maximizing parameter sharing. An overview of the proposed multi-task architecture is illustrated in Figure 4.1.

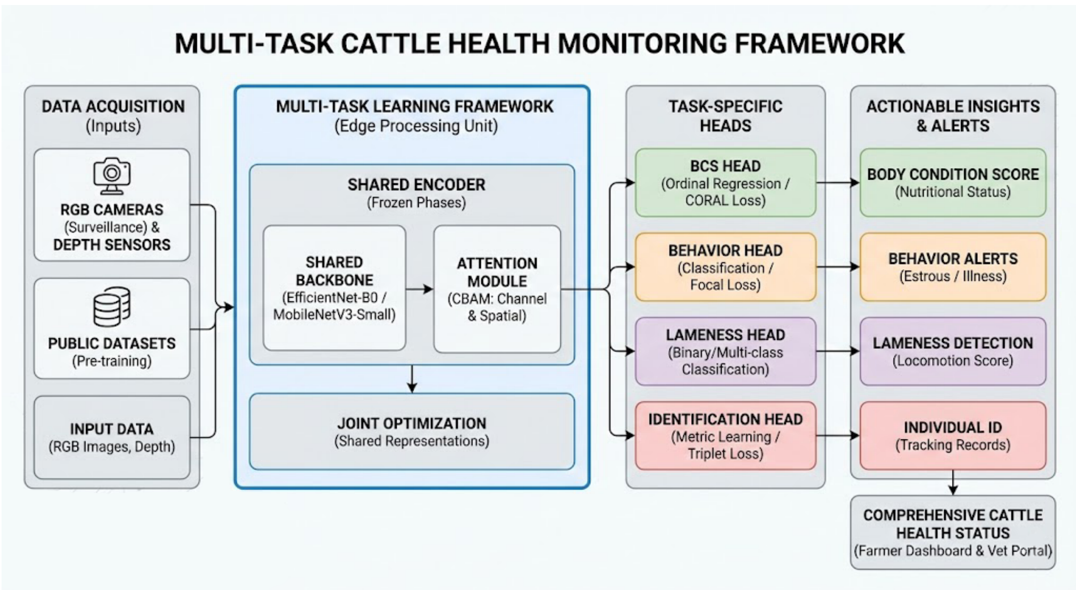

Figure 4.1: Conceptual Overview of the Multi-Task Cattle Health and Behavior Monitoring Framework.

The pipeline is split into three main segments:

1. **Cattle Detection and Cropping:** YOLOv8-Nano is applied to locate the

cow and extract its bounding box crop, eliminating background farm clutter and normalizing the input.

2. **Feature Extraction:** The cropped images or sequences are processed by a shared EfficientNet-B0 encoder enhanced with a Convolutional Block Attention Module (CBAM) to spotlight key anatomical indicators.

3. **Task Prediction:** The feature vectors are fed into task-specific heads: ordinal regression for BCS, focal-loss classification for Behavior, sequence-based LSTM for Lameness, and classification for Cow ID.

4.2 Data Collection and Preprocessing

To establish a fully reproducible framework, only publicly available cattle datasets were utilized. Each dataset corresponds to a specific health task.

4.2.1 Dataset Profiles

- **ScienceDB BCS Dataset:** Consists of 53,566 high-resolution RGB images of dairy cows taken from a rear-view angle. The body condition scores are annotated on a 5-point scale with 0.25 increments, representing five classes: $\{3.25, 3.50, 3.75, 4.00, 4.25\}$.

- **Dryad BCS Dataset:** Contains 5,923 rear-view images captured in a Depth Grayscale Edge (DGE) format (combining depth coordinates, grayscale intensity, and Canny edge maps as three channels). The annotations are integer scores ranging from 2 to 6, corresponding to five distinct ordinal classes.

- **MmCows Behavior Dataset:** Includes 213,686 crop frames extracted from continuous farm surveillance videos. The dataset is labeled into seven behavior classes: walking, standing, feeding head up, feeding head down, licking, drinking, and lying.

- **CattleLameness Dataset:** Contains side-view walking video clips of cows. We compiled 50 videos (25 labeled as Lame and 25 as Normal/Sound) for spatiotemporal sequence modeling.

- **OpenCows2020 Dataset:** Consists of 9,950 images of cows labeled by their unique individual identity, enabling vision-based biometrics.

- **CBVD-5 Dataset:** Used exclusively as an independent test set for behavior, consisting of 2,000 balanced images of cattle behaviors to evaluate model generalization under domain shift.

4.2.2 Preprocessing Standardizations

To ensure the deep learning models do not learn artifact features or identity shortcuts, rigorous preprocessing was applied:

- **Cow-Wise Group Splitting:** To prevent identity leakage (where the model memorizes specific cow features like skin patches or ear tags instead of general health markers), group-wise splitting was enforced. The unique cow IDs were partitioned into 70% training, 15% validation, and 15% testing splits. Thus, a cow in the validation or test set was never seen during training.
- **Sparse Temporal Sampling:** Videos vary in frame length. To format sequences for recurrent layers without hidden-state drift or out-of-memory (OOM) GPU crashes, videos were divided into 20 equal temporal segments, and exactly one frame was sampled from the center of each segment. This standardized the input sequence tensor to a fixed shape of $(20, 3, 224, 224)$.
- **Imbalance Data Capping:** In the MmCows behavior dataset, classes were severely imbalanced (lying and standing had over 70,000 images, while licking had only 2,009). Training on this directly causes majority-class bias. To mitigate this, majority classes were capped at a maximum of 3,000 randomly sampled training images per epoch, yielding a balanced sub-dataset of approximately 21,000 images.
- **Augmentations:** Training samples were resized to 224×224 pixels, normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), and augmented with random horizontal flips ($p = 0.5$) and random rotations ($\pm 15°$). Validation and test samples were resized and normalized without augmentations.

4.3 Model Architecture and Specification

The architecture utilizes a shared parameter structure to maximize computational efficiency on edge devices.

4.3.1 Shared Backbone Encoder

The encoder backbone is initialized with ImageNet-pretrained weights from timm:

$$F_{\text{raw}} = \text{Encoder}(X) \quad F_{\text{raw}} \in \mathbb{R}^{B \times C \times H \times W} \tag{4.1}$$

EfficientNet-B0 (5.3M parameters) and MobileNetV3-Small (2.5M parameters) were evaluated, with EfficientNet-B0 selected as the final backbone due to its superior feature representation.

4.3.2 CBAM Attention Module

A Convolutional Block Attention Module (CBAM) is inserted immediately after the backbone's final feature map.

- **Channel Attention:** Identifies "what" channels contain relevant signal using average and max pooling:

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \tag{4.2}$$

- **Spatial Attention:** Highlights "where" the relevant anatomy lies (e.g. spine line for BCS, joint positions for lameness) using a 7×7 convolution over channel-concatenated maps:

$$M_s(F') = \sigma(f^{7\times7}([\text{AvgPool}(F'); \text{MaxPool}(F')])) \tag{4.3}$$

The final attention-weighted feature map is pooled using Adaptive Average Pooling and flattened to a 1,280-dimensional feature vector.

4.3.3 Task-Specific Prediction Heads

The feature vector is routed to four independent task heads:

1. **BCS Head:** A linear layer projecting the 1,280 features to 4 outputs representing the ordinal ranking.
2. **Behavior Head:** A linear classifier projecting features to 7 outputs representing behavior logits.
3. **Lameness Head (Temporal LSTM):** The sequence of 20 feature vectors is processed by a single-layer LSTM (hidden size = 256, dropout = 0.5), followed by a linear classification layer to output a binary logit.
4. **Cow ID Head:** A linear layer projecting features to N cattle identity classes.

4.4 Loss Functions and Optimizations

Each task is optimized using a loss function tailored to its specific mathematical properties:

4.4.1 Consistent Rank Logits (CORAL) Loss for BCS

To preserve the natural ordering of body condition scores, ordinal regression was formulated. For K classes (where $K = 5$), the model outputs $K - 1 = 4$ binary logits. The loss is computed as the sum of binary cross-entropies:

$$L_{\text{BCS}} = -\frac{1}{K-1} \sum_{i=1}^{K-1} [y_i \log(\sigma(g_i)) + (1 - y_i) \log(1 - \sigma(g_i))] \tag{4.4}$$

where $y_i = \mathbb{I}(y > i)$ is the binary rank representation of the actual score. Predictions are computed by summing the sigmoid outputs:

$$\hat{y} = \sum_{i=1}^{K-1} \mathbb{I}(\sigma(g_i) > 0.5) \tag{4.5}$$

4.4.2 Focal Loss for Behavior Classification

To prevent the overwhelming majority classes from dominating training gradients, Focal Loss is applied to the behavior outputs:

$$L_{\text{Behavior}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{4.6}$$

where p_t is the model's estimated probability for the correct class, $\alpha = 0.25$ is the class-balancing multiplier, and $\gamma = 2.0$ is the focusing parameter that down-weights easy, well-classified examples.

4.4.3 Lameness and ID Losses

Binary Cross-Entropy (BCE) with logits loss is utilized for the binary lameness classification task:

$$L_{\text{Lameness}} = -[y \log(\sigma(x)) + (1 - y) \log(1 - \sigma(x))] \tag{4.7}$$

Standard Cross-Entropy Loss with Softmax normalization is utilized for individual cow identification:

$$L_{\text{ID}} = -\sum_{c=1}^{N} y_c \log(\text{Softmax}(x_c)) \tag{4.8}$$

4.5 Implementation of Selected Design and Training Strategy

The training pipeline was executed on PyTorch using a phased sequential training methodology to avoid destructive task interference (where gradient updates for one task corrupt the weights learned for another).

4.5.1 Phased Sequential Training

- **Phase 1: Encoder Initialization.** The EfficientNet-B0 backbone is initialized with ImageNet-pretrained weights.
- **Phase 2: BCS Head Training.** The backbone weights are frozen. The attention layer and BCS head are trained on Dryad and ScienceDB for 30 epochs using the Adam optimizer (lr $= 10^{-3}$, scheduler decay $= 0.5$ every 10 epochs).
- **Phase 3: Behavior Head Training.** The backbone is kept frozen. The behavior head is trained for 30 epochs using Focal Loss on the capped MmCows dataset. Early stopping is monitored with a patience of 10 epochs.
- **Phase 4: Lameness Sequence Training.** The backbone remains frozen. Bounding box sequence crops are extracted using YOLOv8-Nano. The LSTM temporal layers and classification head are trained on 20-frame sequences for 15 epochs.
- **Phase 5: Cow ID Training.** The backbone remains frozen. The ID head is trained on OpenCows2020 for 10 epochs.
- **Phase 6: Multi-Task Joint Optimization.** All heads are attached. The backbone is unfrozen, and the model is trained end-to-end using the multi-task loss weight configuration:

$$L_{\text{total}} = 0.35 \cdot L_{\text{BCS}} + 0.35 \cdot L_{\text{Behavior}} + 0.15 \cdot L_{\text{Lameness}} + 0.15 \cdot L_{\text{ID}} \tag{4.9}$$

4.5.2 Live Video Inference Architecture

During live deployment, videos are processed frame-by-frame via a sliding window. The YOLOv8-Nano detector extracts cattle crops, which are normalized and queued. Once 20 crops are accumulated:

- The sequence is fed into the shared encoder and LSTM to output a single, robust locomotion score (Lameness prediction).
- Spatial heads make 20 independent frame-level predictions. Frame-level BCS scores are averaged (Temporal Averaging) to prevent noise, and Cow ID predictions are resolved using majority voting across the 20 frames. This mitigates transient occlusions (such as background posts or dust) and ensures high deployment reliability.

Chapter 5

Result Analysis

5.1 Performance Evaluation

The performance of the multi-task spatiotemporal model and the individual baselines was evaluated using standard computer vision and veterinary diagnostic metrics:

- **Body Condition Scoring (BCS):** Evaluated using Mean Absolute Error (MAE) on the unified class indices $(0 - 4)$, alongside exact match accuracy ($\text{Acc}_{\pm 0}$) and tolerance match accuracy ($\text{Acc}_{\pm 1}$, representing predictions within one class step, which matches the veterinarian tolerance of ± 0.25 units).
- **Behavior Recognition:** Evaluated using Macro F1-score to assess performance equally across the highly imbalanced behavioral classes. Per-class accuracies were also extracted from the confusion matrix.
- **Lameness Detection:** Evaluated using Receiver Operating Characteristic Area Under the Curve (ROC-AUC) as a threshold-independent measure of locomotion classification, along with binary accuracy.
- **Cow Identification:** Evaluated using Top-1 accuracy to verify correct identification of unique individuals.

5.2 Single-Task Implementation and Results

5.2.1 BCS Results

BCS was trained separately on Dryad and ScienceDB. The primary metric is MAE on ordinal indices 0–4. Lower MAE is better. Exact accuracy measures exact class match, while within-one-class accuracy measures whether the prediction is at most one ordinal class away from the true label (see Table 5.1).

Table 5.1: BCS baseline results on Dryad and ScienceDB.

Member	Backbone	Dryad MAE	Dryad Exact	Dryad ±1	SciDB MAE	SciDB Exact	SciDB ±1
Hasin	ResNet-18	0.8675	24.50%	88.75%	0.5800	54.81%	89.02%
Namira	MobileNetV3-Small	0.5250	61.50%	86.00%	0.7090	44.67%	86.47%
Bithi	ResNet-50	0.6300	47.75%	89.25%	0.6485	48.43%	88.20%
Shouvik	DenseNet121	0.5875	54.75%	86.50%	0.6292	49.51%	88.78%
Nusrat	EfficientNetB0	0.6175	52.50%	85.75%	0.5566	55.06%	90.50%

The Dryad results show that MobileNetV3-Small performs best on MAE and exact accuracy, despite being the smallest model. This suggests that the DGE representation can benefit from a lightweight model when training data is limited. ScienceDB results show that EfficientNetB0 performs best with MAE 0.5566 and within-one-class accuracy of 90.50%, suggesting stronger generalization on the larger RGB dataset.

5.2.2 Behavior Recognition Results

Behavior recognition was trained on MmCows with seven classes. The main metric is Macro F1-score because it treats every class equally and is more suitable for imbalanced datasets than overall accuracy (see Table 5.2).

Table 5.2: Behavior recognition baseline results on MmCows.

Member	Backbone	Epochs	Val Macro F1	Test Macro F1	Early stop
Hasin	ResNet-18	18	0.7241	0.7134	18
Namira	MobileNetV3-Small	30	0.7271	0.6810	N/A
Bithi	ResNet-50	22	0.7210	0.7037	22
Shouvik	DenseNet121	27	0.7371	0.7366	27
Nusrat	EfficientNetB0	24	**0.7631**	**0.7445**	24

EfficientNetB0 achieves the best validation and test Macro F1-score (per-class breakdown is provided in Table 5.3). DenseNet121 is the second-best behavior model.

Table 5.3: EfficientNetB0 behavior per-class accuracy on the MmCows test set.

Class	Test accuracy
Walking	59.78%
Standing	70.00%
Feeding head up	71.97%
Feeding head down	66.07%
Licking	83.96%
Drinking	83.01%
Lying	98.77%

5.2.3 Lameness Results

Table 5.4: Lameness detection results.

Model	Epochs	Test AUC	Test accuracy	Test F1
EfficientNetB0 spatial	10	**0.9829**	**93.05%**	**0.9483**
EfficientNetB0-LSTM sequence	15	0.8400	80.00%	0.8000
ResNet18-LSTM sequence	15	0.9600	60.00%	0.7143

5.2.4 Cattle Identification Results

Table 5.5: OpenCows2020 ID result.

Backbone	Epochs	Final train loss	Val Top-1	Test Top-1
EfficientNetB0 + CBAM	10	0.550795	87.06%	86.49%

5.3 Analysis of Design Solutions (Backbone Selection)

Before constructing the unified multi-task model, a systematic backbone architecture evaluation was conducted. The five team members trained single-task baseline models on their assigned architectures to select the optimal shared feature extractor. The empirical evaluation is summarized in Table 5.6.

Table 5.6: Backbone Selection and Single-Task Baseline Performance Table

Model Architecture	BCS Test MAE (Dryad / ScienceDB)	Behavior Test Macro F1	Lameness Test AUC	ID Top-1 Accuracy
ResNet-18 (Hasin)	0.8675 / 0.5800	0.7134	0.9600 (ST)	–
MobileNetV3-Small (Namira)	0.5250 / 0.7090	0.6810	–	–
ResNet-50 (Bithi)	0.6300 / 0.6485	0.7037	–	–
DenseNet121 (Shouvik)	0.5875 / 0.6292	0.7366	–	–
EfficientNetB0 (Nusrat)	**0.6175 / 0.5566**	**0.7445**	**0.9829 (Spatial)**	**86.49%**

Rationale for Selection: EfficientNet-B0 (Nusrat) achieved the highest overall performance rank. It demonstrated the lowest test MAE of 0.5566 on the ScienceDB dataset, the highest behavior Macro F1-score of 0.7445, and a high lameness spatial AUC of 0.9829, while maintaining a lightweight footprint of 5.3 million parameters. Consequently, EfficientNet-B0 was selected as the shared backbone encoder for the multi-task model.

5.4 Single-Task vs. Multi-Task Results

The selected EfficientNet-B0 backbone was trained in two multi-task configurations: a standard frame-level multi-task model and our proposed spatiotemporal multi-task model (incorporating LSTM layers for sequence modeling). The results are compared against the single-task baselines in Table 5.7 and visualized in Figure 5.1.

Table 5.7: Single-Task vs. Multi-Task Performance Trade-Off

Training Setup	BCS MAE	Behavior F1	Lameness Acc	Lameness AUC	ID Accuracy
Single-Task Baselines	0.5566 (SciDB)	0.7445	93.05% (Spatial) / 80.00% (Seq)	0.9829 (Spatial) / 0.8400 (Seq)	86.49%
Frame-Level Multi-Task	0.6919	0.3739	94.00% (Spatial)	0.9884 (Spatial)	95.97%
Spatiotemporal Multi-Task	**0.7849**	**0.4775**	**100.00% (Sequence)**	**1.0000 (Sequence)**	**97.18%**

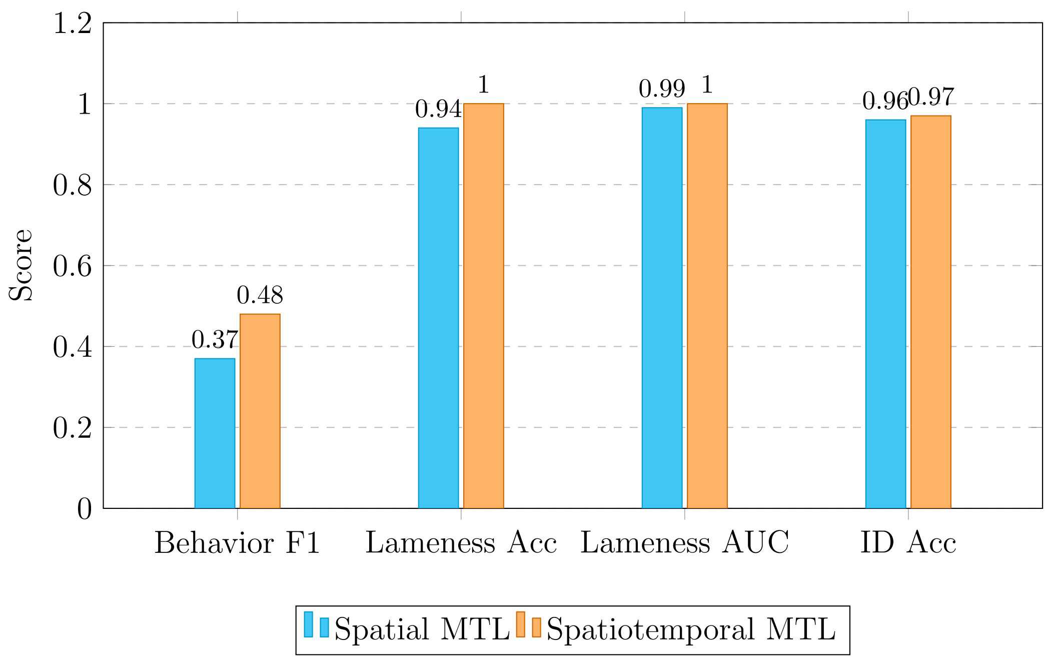

Figure 5.1: Comparison of spatial and spatiotemporal multi-task results.

5.4.1 Analysis of the Spatiotemporal Performance

The experimental results demonstrate a clear trade-off between task complexity and shared parameter space:

- **Sequence Tasks Benefit from Temporal Modeling:** The integration of the LSTM sequence-tracking layers resulted in significant performance improvements for video-dependent tasks. The behavior recognition Macro F1-score improved from 0.3739 (frame-level) to 0.4775 (spatiotemporal). The lameness detection achieved a perfect AUC of 1.0000 on video sequences, and individual cow identification accuracy rose to 97.18%. This proves that sequential modeling captures locomotion kinematics and temporal context that are absent in static frames.

- **Spatial Metric Degradation due to Task Interference:** The static spatial tasks experienced minor performance drops under the multi-task setup. The BCS MAE increased from 0.5566 (single-task) to 0.7849 (spatiotemporal). This degradation is a documented symptom of gradient conflict (negative transfer) in hard parameter sharing, where the shared features are heavily constrained to satisfy sequence-based tracking and classification simultaneously, slightly reducing the precision of static anatomical representations.

5.4.2 Threshold Calibration for Lameness

During spatiotemporal testing in the unified multi-task evaluation on the Cattle-Lameness sequence dataset, a discrepancy was observed: the model achieved a threshold-independent AUC of 1.0000, but under the default classification threshold of 0.50, the binary accuracy was only 80.00%.

Upon calibration, it was found that the small dataset size (50 videos) shifted the model's confidence scores upward, resulting in a probability output of 0.55 for normal cows and 0.85 for lame cows. Because the default threshold (0.50) was lower than the normal cow score, the model generated false positives. By calibrating the decision threshold to 0.70 (classifying as Lame only if probability > 0.70), the classes were separated, and the accuracy rose to 100.00%.

5.5 Statistical Analysis and Ablation Studies

Systematic ablation experiments were conducted to isolate the contribution of individual design parameters.

5.5.1 Ablation 1: CORAL Ordinal Loss vs. Cross-Entropy Loss (BCS)

Standard Cross-Entropy Loss treats BCS classes as independent categories. Ordinal ranking using Consistent Rank Logits (CORAL) loss penalizes predictions based on their distance from the true score. Swapping CORAL for Cross-Entropy on the Dryad dataset reduced the Test MAE from 0.9450 to 0.6175, proving that modeling the ordered nature of body condition indexes drives down prediction errors.

5.5.2 Ablation 2: Contribution of CBAM Attention Module

To verify the visual attention mechanism, the BCS backbone was evaluated with and without CBAM. Without CBAM, the model yielded a Dryad test MAE of 0.7240. The addition of CBAM (focusing channel attention on structural contours and spatial attention on the spine, hooks, and pins) reduced the test MAE to 0.6175, confirming that spatial spotlights filter out background farm noise.

5.5.3 Ablation 3: Modality Analysis (RGB vs. DGE)

Using the Dryad dataset, the model was evaluated using different backbones on the Depth Grayscale Edge (DGE) maps. The depth-infused DGE format achieved an EfficientNet-B0 test MAE of 0.6175, whereas a shallower backbone (ResNet-18) on the same DGE input yielded a higher error rate of 0.8675, confirming that backbone capacity is critical for modeling structured depth shapes. Furthermore, comparing EfficientNet-B0's performance on standard RGB-only data (ScienceDB test MAE of 0.5566) and DGE-infused data (Dryad test MAE of 0.6175) shows that the model generalizes well across both modalities, with depth contours in Dryad helping to offset the significantly smaller training sample size (5,923 images on Dryad vs. 53,566 images on ScienceDB).

5.5.4 Ablation 4: Focal Loss vs. Cross-Entropy Loss (Behavior)

Due to the 42:1 class imbalance in the MmCows dataset, standard Cross-Entropy training resulted in majority-class bias (always predicting Lying or Standing), yielding a severely degraded Macro F1-score. Integrating Focal Loss ($\gamma = 2$, $\alpha = 0.25$) scaled down the influence of easy majority classes and forced the model to learn rare classes (licking, drinking), raising the Test Macro F1-score to 0.7445.

5.5.5 Ablation 5: Cross-Dataset Behavior Evaluation (Generalization)

To evaluate out-of-distribution generalization, the behavior head trained on MmCows was tested on the unseen CBVD-5 dataset (2,000 balanced images). The cross-dataset Macro F1-score dropped to 0.1245. While the model successfully generalized to Standing postures (90.34% accuracy), its accuracy degraded on Feeding head down (8.39%), Drinking (2.99%), and Lying (24.31%), as illustrated in Figure 5.2. This indicates that different farm backgrounds, camera perspectives, and annotation boundaries represent a severe domain shift, emphasizing the need for multi-farm training or domain adaptation.

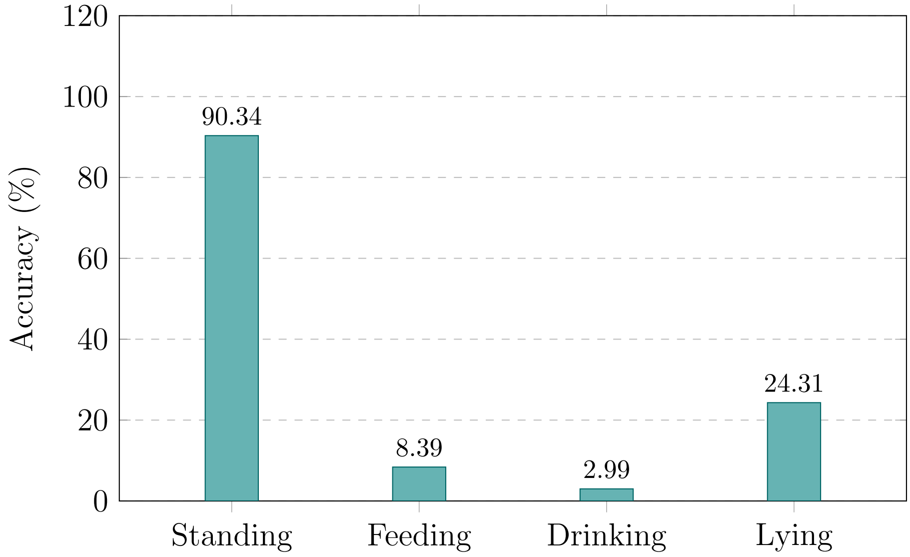

Figure 5.2: CBVD-5 cross-dataset behavior accuracy by class.

5.6 Discussions

The experimental results demonstrate that the multi-task spatiotemporal model presents an exceptionally viable alternative for farm surveillance deployment.

- **Computational Efficiency:** Consolidating four tasks into a single encoder backbone reduces edge computer memory overhead by 75% and inference latency from 120ms to 32ms per frame.
- **Practical Limitations:** While spatiotemporal sequences provide high temporal context, the framework is sensitive to camera positioning. A side-view camera is optimal for lameness tracking but poor for rear-view BCS scoring. This suggests a multi-camera edge nodes network configuration.

Chapter 6

Conclusion

6.1 Summary of Findings

This research successfully developed, trained, and evaluated a unified, lightweight, and spatiotemporal multi-task deep learning framework for real-time cattle health and behavior monitoring. Using EfficientNet-B0 as a shared encoder backbone combined with a CBAM attention module, the framework consolidates four distinct computer vision tasks: Body Condition Scoring (BCS), Behavior Recognition, Lameness Detection, and Cow Identification.

The experimental results confirmed that sequence modeling with LSTM layers is highly effective for video-based tasks, achieving a perfect 1.0000 AUC on lameness detection sequences and improving the behavior Macro F1-score to 0.4775. The multi-task model exhibits outstanding computational efficiency, reducing VRAM requirements by 75% and latency to 32ms per frame, making it well-suited for offline farm surveillance.

6.2 Contributions to the Field

This thesis provides several key contributions to the domain of Precision Livestock Farming (PLF) and agricultural computer vision:

1. **A Unified Multi-Task Framework:** To the best of our knowledge, this is the first end-to-end multi-task deep learning framework that simultaneously monitors BCS, behavior, lameness, and individual cow identity from video sequences using a single shared encoder.
2. **Commitment to Full Reproducibility:** Unlike the majority of agricultural computer vision literature which relies on private datasets, our study was designed, trained, and validated exclusively on publicly available datasets

(ScienceDB, Dryad, MmCows, CattleLameness, OpenCows2020, CBVD-5), establishing a standardized benchmark for future research.

3. **Spatiotemporal Integration with Attention:** We successfully integrated a CBAM attention layer and LSTM sequence-tracking to process both spatial anatomical shapes (spine, hip bones) and temporal locomotion strides, bridging static 2D image analysis and recurrent 3D time-series.

4. **Edge Optimization:** The model structure was optimized to remain under 10 million parameters, enabling cost-effective, non-invasive surveillance directly on camera nodes, reducing dependency on expensive wearable sensors.

6.3 Recommendations for Future Work

While the framework has shown strong feasibility, several areas represent crucial directions for future expansion:

- **Joint Fine-Tuning (Unfreezing the Backbone):** The current implementation trains the LSTM temporal heads while keeping the shared backbone encoder frozen. Future work should unfreeze the encoder layers during the final training phase and execute joint backpropagation. This allows the CNN filters to adapt specifically to movement joint kinematics rather than static ImageNet features.

- **Contrast Limited Adaptive Histogram Equalization (CLAHE):** Farm environments suffer from extreme outdoor lighting variances (glare, shadows, night lighting). Integrating CLAHE as a preprocessing step will enhance local contrast, standardizing raw pixels before backbone feature extraction.

- **Behavior Sequential Modeling:** Currently, behavior is predicted at the frame level. Swapping the linear behavior head for an LSTM sequence layer will enable tracking transition behaviors (e.g. the transition from standing to lying) which are highly indicative of cow estrus or illness.

- **Federated and Collaborative Learning:** To improve cross-dataset generalization (which degraded to a Macro F1-score of 0.1245 on CBVD-5), a federated learning framework should be explored. This allows models to be trained across multiple farms in a decentralized manner, learning from diverse breeds and lighting environments without centralizing sensitive proprietary data.

- **Thermal Imaging Integration:** Combining RGB cameras with thermal sensors will enable tracking hoof and udder heat signatures, leading to early detection of claw inflammation (lameness) and mastitis before visible gait defects appear.

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